MOCA — Template Onboarding Guide

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This guide takes a container image to a **sandbox template**: a named image that a client, or an agent session, asks for by id, and whose commands then run in microVMs restored from that image's snapshot. It walks through two worked examples end to end: MOCA's public sandbox image (anyone can reproduce it) and a SWE-bench instance (the benchmark case it was built for). It was validated on 2026-10-29 and 2026-10-30 against an SBX0 deployment: the control plane and relay on OpenShift, and the workers on a bare-metal KVM host.

- **A template is one worker serving one snapshot.** Onboarding is: build a snapshot from the image, check it in a real microVM, give it a worker, register it, use it, and retire it when a new image replaces it.
- **Everything for templates is in this folder:** the tools, the two sample specs, their tests, and links to the related files elsewhere.
- The short reference for the same steps is `deploy/sbx/template-onboarding/ONBOARDING.md`. Deploying SBX0 itself is `deploy/sbx/README.md` and `RUNBOOK.md`; using a sandbox is the user guide; templates in MOCA's own agent sessions are designed in `2026-10-30-session-sandbox-templates-design.md`.
- **Every host and registry below is a placeholder:** `sbx-api.example.com`, `registry.example.com`. No real host, account id or token appears here.

In a hurry? §2 is the whole flow on one page; §4 is the public-image example as copy-pasteable commands; §6 is a client choosing between the two sample templates.

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1. Mental model (read this first)

1. **A sandbox runs every command in a fresh microVM** restored from its template's golden snapshot, with the sandbox's own `/workspace` attached. Files in `/workspace` persist across commands; nothing else does.
2. **One worker (`microvm-worker`) serves one snapshot.** A template is a name for a worker, so a new image of a template is a new snapshot and a new worker; the old one drains.
3. **The client chooses the template** when it creates a sandbox (`mocactl sbx create --template ID`, `POST /v1/sandboxes {template}`), and a sandbox keeps its worker for life.

What a snapshot holds

Every template is its own bootable image: there is no common MOCA image underneath. What templates share is MOCA's thin boot layer, which `build-snapshot.sh` copies into each snapshot.

In a snapshot	Shared by every template?	Size (a 4 GiB SWE-bench guest)
`kernel`	yes: the host's one guest kernel (`--kernel`)	~28 MB
`agent`	yes: MOCA's guest agent, from `remote-worker`	~3 MB
`/sbin/init`	yes: MOCA's init (mounts, `/dev/fd`, `/var` overlay, `HOME`), inside the rootfs	a few KB
`rootfs`	no: an ext4 image of the template's container image (OS, toolchain, repository, `/var`)	3.5-9 GB
`memfile`, `vmstate`	no: the guest's RAM and CPU state after it booted	the guest's RAM (4 GiB), a few KB

So each template costs 7-14 GB of disk, mostly its rootfs and memfile, and disk is what limits how many templates a host holds (§9). Within one template every VM is restored from the same files by hardlink and gets only its own workspace disk, so more sandboxes of a template cost little.

Where things run

Component	Runs on	Does
`template-build.sh`	the KVM host, as root	image → snapshot, checked in a microVM → a template record

Component	Runs on	Does
`template-register.sh`	the operator's machine	record → SBX0's inputs (`SBX_WORKERS`, the templates file)
`setup.sh`	the operator's machine, against the cluster	mints the worker's token and bundle; restarts the control plane only
`install-host.sh`	the KVM host, as root	installs the worker, which attaches to the relay
`mocactl sbx`	anywhere	lists templates, creates sandboxes, runs commands

Names

For a template id `ID` and an image digest `sha256:abcdef012345…`:

What	Name
snapshot	`ID-abcdef012345` (the first 12 hex of the digest)
worker id	`sbx_<ID with - as _>_abcdef012345`
record	`$SBX_SRV/templates/<snapshot>.json`

2. The flow on one page

#	Where	Command	Check
1	anywhere	write `spec.json` (§3)	`template-build.sh` validates every field first
2	KVM host	`template-build.sh spec.json --kernel K [--registry H]`	digest, `/etc/hosts`, the probe in a microVM; a failure publishes nothing
3	operator	`template-register.sh --env-file F record.json`	one worker per template; an update keeps the old worker
4	operator	`setup.sh --env-file F`	the control plane rolls; the relay does not
5	KVM host	`install-host.sh BUNDLE --snapshot S --budget-mb N [--no-egress]`	the worker attaches
6	client	`mocactl sbx create --template ID`	the sandbox runs on that template's image
7	both	new image: steps 1-5 again, then `--retire` the old worker	its live sandboxes have ended

3. The template spec

A JSON file, schema `moca-sbx-template-spec/1`. Every field is checked before anything runs (`template-build.sh`):

Field	Meaning
`id`	what clients ask for: `^[a-z0-9][a-z0-9-]{0,30}$`; the same across versions
`image`	`repository@sha256:<digest>`, never a tag; a repository with no host takes it from `--registry`
`guest_ram_mb`, `vcpus`	the guest's size, fixed in the snapshot

Field	Meaning
`guest_home`	optional, a writable `HOME` under `/tmp` (the rootfs is read-only)
`skip_git_identity_check`	`true` for an image that refuses agent commits on purpose
`guest_home_gitconfig`	optional, `true`: copy the image's `/root/.gitconfig` into `guest_home`; refused with `skip_git_identity_check`
`egress`	`false`: no network at all (`--no-egress`); `true`: the host's allowlisted egress
`seed`	a bash command that fills a new sandbox's empty `/workspace`, or `""`
`probe`	`{cmd, timeout_s, expect_rc, expect[]}`: run after the seed in the restored VM; passes on the exit code and every regex
`description`, `tools`	what clients see in `GET /v1/templates`
`etc_hosts_sha256`	optional: the build stops unless the rootfs's `/etc/hosts` has this hash

The probe must be shareable

The probe proves the image's toolchain works in a microVM, not that a task is solvable. For a benchmark, **use a public test target, never the tests that grade a fix**: their names and counts are answer-key material. Match the runner's output as it prints it (§10 has two ways this went wrong).

4. Example 1: MOCA's public sandbox image

MOCA's own sandbox image, `ghcr.io/rossoctl/moca-remote-worker`, is public, so this example needs no registry key. It is the image SBX0's `default` template ran on 2026-10-29 (built then with `build-snapshot.sh` directly). On 2026-10-31 this exact spec was built with `template-build.sh` (with `--egress-ca`, for egress credentials) and its probe passed in the restored VM.

4.1 Write the spec

This spec is `examples/default.json` in this folder:

```
{
  "schema": "moca-sbx-template-spec/1",
  "id": "default",
  "image": "ghcr.io/rossoctl/moca-remote-
   worker@sha256:eed901b838bf11e16054fccfb38af600a6e61d74dd52907af0750bba13ecbed3",
  "guest_ram_mb": 256,
  "vcpus": 1,
  "egress": true,
  "seed": "",
  "probe": {
    "cmd": "git --version && python3 -c 'print(6*7)' && rg --version | head -1",
    "timeout_s": 120,
    "expect_rc": 0,
    "expect": ["^git version ", "^42$", "^ripgrep "]
  },
  "description": "MOCA's sandbox image: bash, git, Python 3, ripgrep",
  "tools": ["bash", "git", "python3", "rg"]
}
```

`egress: true` gives the guest the host's allowlisted egress (GitHub by default, RUNBOOK). Use `false` for a template that must stay offline.

4.2 Build and probe on the KVM host

```
sudo deploy/sbx/template-onboarding/template-build.sh deploy/sbx/template-
   onboarding/examples/default.json --kernel /path/to/vmlinux
```

Expected output, ending:

```
build-snapshot.sh: verify command exited 0; the last of its output:
...
template-build: the probe passed
==> built template default: snapshot default-eed901b838bf, worker sbx_default_eed901b838bf
==> record: /srv/sbx/templates/default-eed901b838bf.json (copy it to the operator's machine for
   deploy/sbx/template-onboarding/template-register.sh)
==> then, after setup.sh: sudo deploy/sbx/install-host.sh <bundle of sbx_default_eed901b838bf> -
   -snapshot default-eed901b838bf --budget-mb <at least 1888>
```

A 256 MiB guest's snapshot is small; a 4 GiB one is about 7.3 GB (§9).

4.3 Register (operator machine)

Copy the record over, then:

```
deploy/sbx/template-onboarding/template-register.sh --env-file <env file> default-
   eed901b838bf.json --apply
```

It adds the worker to `SBX_WORKERS` and the template to the templates file, then runs `setup.sh`. Only the control plane restarts, so running commands on other templates are not interrupted (on 2026-10-30 the relay's settings hash stayed `1cf8f5032ae16280` across two template changes).

4.4 Install the worker

Copy the bundle `deploy/sbx/.generated/workers/sbx_default_eed901b838bf` to the host by path (it holds a token: never print it), then:

```
sudo deploy/sbx/install-host.sh <bundle dir> --snapshot default-eed901b838bf --budget-mb 8192
```

Expected: `sbx-worker@sbx_default_eed901b838bf attached to the relay`. The worker first checks the snapshot against its manifest and against the template record `template-build.sh` left in `$SBX_SRV/templates/` (`SH_TEMPLATE_RECORD`); a mismatch stops it at start, naming the field. The record's id also becomes the worker's `SANDBOX_TEMPLATE`, so MOCA agent sessions naming the template find it (section 8).

4.5 Use it

```
mocactl sbx templates
mocactl sbx create --template default --name try-default
mocactl sbx run -- python3 -c 'print(6*7)'
```

5. Example 2: a SWE-bench instance (django__django-11133)

This is exactly what ran on 2026-10-30 (the record is in the project's run notes): a SWE-bench Lite instance, built, checked in a microVM, registered, used by a client and by an agent, then retired.

5.1 The spec comes from image-factory

The result, as it ran, is `examples/django__django-11133.json` in this folder (its image names no registry host; give it with `--registry`).

image-factory's moca-microvm records carry the spec under `template`, so nothing is written by hand. Its `probe` is a **public** test target (`basic`), not the instance's grading tests, and its image names no registry host:

```
jq '.template' platforms/moca/microvm/django__django-11133.json > django__django-11133.json
cp platforms/moca/microvm/django__django-11133.json django__django-11133.gate-record.json
```

The second file is the image's **gate record**, unchanged: the build checks the image against it (§5.2). Both are in `examples/`.

The fields that matter: `id: django-django-11133` (the instance id lower-cased, `__` → `-`; a Verified image also in Lite gets `-v`), 4096 MiB and 2 vCPUs, `egress: false`, `seed: "cp -a /opt/testbed-src/. /workspace/"`, `probe.expect: ["^OK\\b"]` with `expect_rc: 0`, and `etc_hosts_sha256`.

5.2 Build, register, install

```
sudo deploy/sbx/template-onboarding/template-build.sh deploy/sbx/template-
   onboarding/examples/django__django-11133.json --kernel /path/to/vmlinux --registry
   registry.example.com --gate-record deploy/sbx/template-onboarding/examples/django__django-
   11133.gate-record.json
```

The spec names its gate record (`source_record_digest`), so the build needs it: before pulling, it checks the gate passed, the record is for this image, and commits are refused, and the template record keeps that evidence (gating rules X10, X12). The run below was on 2026-10-30, before this check existed *(the `--gate-record` form is unverified on a host)*.

As it ran on 2026-10-30 (about 3 minutes, plus the pull):

```
==> etc/hosts as the spec says
build-snapshot.sh: verify command exited 0; the last of its output:
Ran 53 tests in 0.062s
OK (skipped=2)
template-build: the probe passed
==> built template django-django-11133: snapshot django-django-11133-3d3e705ff379, worker
   sbx_django_django_11133_3d3e705ff379
```

Then §4.3 and §4.4 with that record, and **`--no-egress` with `--budget-mb 13408`**: a 4 GiB guest needs (4096 + 32) × 3 + 1024 MiB for one session at standby depth 2.

5.3 A client: seed first

A new sandbox's `/workspace` is empty. **SBX0 does not run the seed: the client does**, as its first command. And use `sbx create`, not `sbx up`: `up` pushes the client's own repository into `/workspace`

before the seed runs.

```
id=$(mocactl sbx create --template django-django-11133 --json | jq -r .id)
mocactl sbx run --sandbox "$id" -- bash -c 'cp -a /opt/testbed-src/. /workspace/ && git -C
   /workspace log --oneline -1'
mocactl sbx run --sandbox "$id" -- bash -c 'source /opt/miniconda3/bin/activate && conda
   activate testbed && cd /testbed && ./tests/runtests.py --verbosity 2 --settings=test_sqlite -
   -parallel 1 basic 2>&1 | tail -3'
```

Expected (2026-10-30): the seed in 1-2 s with a clean tree at `bc34802 baseline`, then `Ran 53 tests`, `OK (skipped=2)`. The guest has no network (`connect: Network is unreachable`), and a token for this sandbox is `PERMISSION_DENIED` at any other template's worker.

5.4 An agent: OpenCode on the template

Point the agent's MCP server at the seeded sandbox (`mocactl sbx mcp --sandbox <id>`), keeping `sbx up`'s tools-off block, so the agent works only through the sandbox. On 2026-10-30 a headless `opencode run` with the instance's problem statement used `sandbox_bash`, `sandbox_read` and `sandbox_edit` only, finished in 23 s, and its patch was **graded resolved** by the official SWE-bench harness. Take the patch from the sandbox (`git diff --binary <baseline>` in `/workspace`), not with `sbx pull`, which diffs against the client's repository.

6. Two templates side by side: the client chooses

With both examples onboarded, the deployment serves two templates, each on its own worker. The templates file that `template-register.sh` wrote holds:

```
[
  {
    "id": "default",
    "worker": "sbx_default_eed901b838bf",
    "description": "MOCA's sandbox image: bash, git, Python 3, ripgrep",
    "tools": ["bash", "git", "python3", "rg"]
  },
  {
    "id": "django-django-11133",
    "worker": "sbx_django_django_11133_3d3e705ff379",
    "description": "SWE-bench django__django-11133 (lite) at its baseline bc348023edb4; seed
   with: cp -a /opt/testbed-src/. /workspace/",
    "tools": ["bash", "git", "python3", "rg"]
  }
]
```

A client lists them, then picks one per sandbox. The worker ids never reach the client: it sees only the id, the description and the tools.

```
mocactl sbx templates
```

```
default          MOCA's sandbox image: bash, git, Python 3, ripgrep
django-django-11133 SWE-bench django__django-11133 (lite) at its baseline bc348023edb4; seed
   with: cp -a /opt/testbed-src/. /workspace/
```

(`--json` gives the same list as JSON, with `tools`.)

A sandbox of each, and a command that tells them apart (only the SWE-bench image has the instance's repository at `/opt/testbed-src`):

```
gen=$(mocactl sbx create --template default --name general --json | jq -r .id)
swe=$(mocactl sbx create --template django-django-11133 --name task --json | jq -r .id)
for s in "$gen" "$swe"; do
  mocactl sbx run --sandbox "$s" -- bash -c 'git -C /opt/testbed-src log --oneline -1
   2>/dev/null || echo no-testbed'
done
```

Expected:

```
no-testbed
bc34802 baseline
```

The SWE-bench template also has no network (`egress: false`, checked on 2026-10-30: `connect: Network is unreachable`). `default`'s egress (`egress: true`, through the host's allowlisting proxy, so `git` and `curl` over HTTPS rather than raw TCP) was checked on 2026-10-31 with SBX0's egress credentials: a second user's sandbox cloned, committed and pushed a branch of a private repository with its stored PAT, which never entered the sandbox; another user's sandbox got `401`; deleting the credential revoked it within its cache TTL (`smoke.sh` step 7).

What SBX0 guarantees, as checked live with two templates on 2026-10-29 (`deploy/sbx/templates-check.sh`, 7 checks, 0 failed):

- the list names every template and **no worker id**;
- each create records the template asked for; an **unknown template is refused** (`no such template`) and creates nothing;
- each template's sandboxes go to **its own worker**; two sandboxes of one template share the worker but **not their workspaces**;
- each sandbox's commands run on **its template's image** (the probe saw each image's digest);
- a token for one template's sandbox sent to the other template's worker is **`PERMISSION_DENIED`** at the relay;
- another user cannot see or use the sandbox;
- `sbx up --template` refuses a repository already bound to a sandbox of another template.

The 2026-10-29 run used `default` and an earlier SWE-bench template, `swe-astropy-14995`; the 2026-10-30 run repeated the choice and isolation checks with `django-django-11133` and `astropy-astropy-14995`. To check a deployment yourself, run `templates-check.sh` against two of its templates (RUNBOOK §14).

7. Update and retire

A new image for a template: a new spec with the new digest, then build, register and install again. The record names a new worker; registering moves the template there and **keeps the old worker** so its live sandboxes keep working. Once none is left:

```
deploy/sbx/template-onboarding/template-register.sh --env-file <env file> --retire <old worker
   id> --apply
sudo deploy/sbx/install-host.sh --uninstall <old worker id>
```

Removing a template entirely is `--remove ID`. On 2026-10-30 retiring `django-django-11133` revoked its worker's token, a create of it was refused (`no such template`), and an astropy sandbox created before kept running.

Self-serve, through the API (a deployment with the template store and host agents; deployed and shown end to end on 2026-11-06): the template's owner does it all, with no operator step.

Point `default` at a new build (new creates use it), then retire the old build, or the whole template:

```
mocactl sbx templates tag django-django-11133 default --build sha256:<new build>
mocactl sbx templates retire django-django-11133 --build sha256:<old build>
mocactl sbx templates retire django-django-11133 --reason 'done'
```

A retire is refused while a sandbox or a session still uses the build (`409 build_in_use`): delete it first. A retired build is never used again (`410 template_retired`), and the host agent then removes its worker and snapshot. Every step is a record naming who did it. The owner also changes the description and visibility (`templates edit`); an admin rejects a build after the fact, transfers a template (`templates transfer`) and sets quotas (`templates quota`).

8. Templates in MOCA agent sessions

MOCA's own agent sessions (Moca Pi, P6) can name a template too: `mocactl run --option sandboxTemplate=ID`. A worker serves a template when `SANDBOX_TEMPLATE` is set; a session with no template never lands on one; the harness runs the seed once, before the agent's first tool call. A templated P4 worker is installed beside the main one with `setup-microvm.sh --template ID --snapshot DIR`; set-up is in `P4-ON-P6.md`. Only one worker per host may use guest egress (moca-egress keys VMs by id, which workers share), so the installer refuses `--egress` beside another worker that uses it. Checked live on a MOCA-VM (P6+P4) on 2026-10-31, with a scripted mock model: a session naming `django-django-11133` ran on the templated worker with the repository seeded, its second turn kept the workspace and did not seed again, a session with no template ran on the untemplated worker, and an unknown template was refused at creation.

On the **combined control plane** (one control plane for both, `setup.sh SBX_PROFILE=combined`), a session names a self-serve template the same way, and may pin a build (`ID@sha256:…`): it runs only on that build's workers. Checked on ykt3 and bm1 on 2026-11-05, with no model: a stock-agent sandbox and a session pinned to the same build, and first use through a session (`409 template_not_active`, then created 23 s later).

9. Capacity

Measured on one host (188 GiB, 93 GB free) on 2026-10-31, with SWE-bench templates of 4 GiB guests:

Limit	Rule	Measured
disk, templates	7.3-13.6 GB per snapshot; `template-build.sh` wants `SBX_MIN_FREE_GB` (50 by default) free	8 built in 31 min (140-515 s each, ~70 GB); room for about one more
memory per template	each attached worker keeps its snapshot's memory mapped, idle or not	~4.1 GiB each
disk, sandboxes	each seeded sandbox's workspace	75-240 MB each; 25 sandboxes over 8 templates and 2 users took ~3.5 GB, and disk stopped the test first
commands at once	`--budget-mb` ≥ (guest + 32) × 3 + 1024; a **running** VM is charged its full guest size, a standby its dirty pages	3 per worker at 13408 MiB; 11 of 11 at 65536 MiB, with almost no real memory used
quotas	`SBX_MAX_PER_SUBJECT`, `SBX_MAX_TOTAL`	deployment settings
logins	a minted login lasts at most 24 h	—

So on a host like this, disk limits templates and sandboxes before memory does, and the default budget allows far fewer commands at once than memory holds: raise `--budget-mb` for parallel runs.

10. Troubleshooting (what we actually hit)

Symptom	Cause	Fix
`no match for ^Ran 51 tests?\b`, though the tests passed	an expected count rebuilt from "passed", but Ran counts skips	expect what the runner prints (`^OK\b`, or the Ran total)
`no match for \b8 passed\b`, though pytest passed	pytest coloured its summary (`ESC[32m8 passed`)	fixed in `template-build.sh`: it matches with ANSI stripped
a probe expectation from the gate's counts	the gate counts include grading tests	use a public target (§3)
`git checkout` fails part way as a non-root user	root-owned files from earlier root builds in the checkout	`chown -R` the checkout to its user first; check with `find … -not -user …`
a worker restarts and fails	its env names a snapshot no longer on disk	point it at one that exists before starting
`sbx up` sandbox has the client's repo, not the template's	`up` pushes the client repository	`sbx create`, then the seed (§5.3)
every restore after `vm-9` fails "API socket never appeared"	the jail path overflowed a Unix socket for a 31-character template id	fixed (2026-10-31): jails are keyed by a short hash of the worker id
"no capacity" for minutes, never ending	the session's tier or template has no worker attached	the retry line (and the harness log) now name the cause; attach a worker of it

11. Regenerating the documents

Artifact	Generated by	Source of truth
this guide's table of contents	`python3 docs/tools/gen_toc.py deploy/sbx/template-onboarding/GUIDE.md`	the headings

Artifact	Generated by	Source of truth
`deploy/sbx/template-onboarding/GUIDE.pdf`	`python3 docs/tools/gen_pdf.py`	this `.md`

If no generator rebuilds it, don't commit it.